ASTRA: An Integrated Analysis Framework for Physics-Aware Astrophysical Discovery

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ABSTRACT

We present ASTRA (Autonomous System for Scientific Discovery in Astrophysics), an integrated framework combining numerical data analysis, causal reasoning, and physical validation. ASTRA integrates: dimensional analysis; structural causal models; Bayesian evidence computation; and multi-wavelength data fusion within a unified framework.

We demonstrate capabilities through six test cases. The first five use real observational data to validate ASTRA's ability to recover known astrophysical results, including: Herschel filament scaling relations with dimensional analysis; Chandra Deep Field South multi-wavelength cross-matching with Bayesian uncertainty propagation; SDSS galaxy property correlations; Gaia stellar causal inference; and Bayesian model selection for virial scaling relations.

The sixth test case demonstrates discovery-mode operation using synthetic supernova data (3,000 SNe) with known ground truth. Applying ASTRA's discovery enhancement system, we recover the Phillips relation (decline rate–luminosity correlation), establish its causal structure through temporal analysis, and identify high-priority targets. This synthetic test validates that the system can discover known patterns when embedded in realistic data.

Code and documentation: `https://github.com/Tilanthi/ASTRA` White & Dey (2026).

1 INTRODUCTION

Modern astronomical surveys generate petabytes of data requiring automated analysis. Multiple computational approaches are available to astronomers, each with distinct strengths and limitations.

Traditional Machine Learning and Statistical Methods: Well-established tools including scikit-learn, emcee, and nested sampling codes (MultiNest, dynesty) can process numerical data, detect patterns, perform statistical inference, and compute Bayesian evidence. These methods are widely used in astronomy with demonstrated success.

Large Language Models with Code Execution: Current frontier models (GPT-4 with Code Interpreter, Claude with code execution) can process numerical data, execute Python code, and perform statistical analysis. These systems provide conversational interfaces to computational tools but require explicit programming and do not natively integrate physical validation or causal interpretation.

The Integration Challenge: While individual tools exist for specific tasks, astronomers typically must manually combine separate pipelines for data processing, statistical analysis, causal inference, and physical validation. This manual integration creates opportunities for inconsistent treatment of uncertainties, missed physical constraints, and difficulty reproducing multi-step analyses.

ASTRA addresses this challenge by providing an integrated framework that combines:

- Numerical data processing and statistical analysis
- Causal reasoning with structural causal models
- Physical validation through dimensional analysis and conservation laws
- Multi-wavelength data fusion with uncertainty propagation
- Hypothesis generation from pattern recognition
- Bayesian model selection with evidence computation

This integrated approach provides a unified framework where physical validation, causal reasoning, and uncertainty quantification are applied consistently throughout multi-step analyses. While individual components use established algorithms (PC algorithm for causal discovery, bridge sampling for Bayesian evidence, dimensional analysis for physical validation), their integration within a single framework enables analyses that would otherwise require manual combination of multiple separate tools.

In this paper, we focus on six examples demonstrating how this integrated approach can be applied to astrophysical inference problems:

(i) **Scaling Relations Analysis** (Section 3): Dimensional analysis and physical validation of filament scaling relations

(ii) **Multi-Wavelength Data Fusion** (Section 4): Probabilistic cross-matching with Bayesian uncertainty propagation

(iii) **Pattern Recognition** (Section 5): Identification of established galaxy property relationships

(iv) **Causal Inference** (Section 6): Structural causal model discovery in stellar data

(v) **Bayesian Model Selection** (Section 7): Evidence-based model comparison with prior specification

(vi) **Discovery-Mode Operation on Synthetic Data**

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(Section 8): Discovery enhancement system applied to supernova light curves

Tests 1–5 use real observational data, validating ASTRA's ability to correctly recover known astrophysical results. Test 6 uses synthetic supernova data with known ground truth to validate discovery-mode capabilities.

Section 9 discusses why these results demonstrate ASTRA's unique capabilities, clarifies ASTRA's role as a tool to assist rather than replace astronomers, and Section 10 concludes.

1.1 Scope and Limitations

What ASTRA Demonstrates: The six test cases in this paper show ASTRA can: discover physical laws from data with theoretical validation; fuse multi-wavelength observations with proper uncertainty handling; identify established astrophysical relationships through pattern recognition; distinguish causation from correlation; perform rigorous Bayesian model comparison; and operate in discovery mode to detect embedded patterns, establish causal structure, and prioritize targets in synthetic data with known ground truth.

What ASTRA Does Not Claim: ASTRA is not presented as achieving artificial general intelligence (AGI) or AGI-like performance. The system operates within defined astrophysical domains using established algorithms (PC algorithm, Bayesian inference, dimensional analysis, FCI causal discovery) combined through an integrated architecture. Results are validated against known physical theory and observational constraints. The system does not claim general reasoning beyond its training domains or autonomous operation without human oversight. The architecture is designed to support natural language queries that trigger autonomous task completion, though this capability is not demonstrated in the present test cases.

ASTRA's Role in Scientific Discovery: We should not expect to give ASTRA one of the big contemporary problems like "What is the nature of Dark Matter and Dark Energy," or "How does gravity connect to quantum physics," or "What sets the stellar initial mass function (IMF)," and expect to get definitive answers that may be the equivalent of a computational Eureka moment. But ASTRA's discovery architecture, used alongside and by an experienced astronomer, has capabilities in discovery and inference that go beyond those of straightforward AI (large language models) and/or machine learning. ASTRA's role in science is to work alongside the experienced astronomer to analyze data and facilitate genuine discovery. ASTRA is therefore a tool to assist the astronomer, rather than a replacement for domain expertise.

Validation Scope: Tests 1–5 use real observational data to validate ASTRA's ability to correctly recover known astrophysical results. Test 6 uses synthetic supernova data with known ground truth to validate discovery-mode capabilities. The capabilities demonstrated are shown within specific astrophysical domains using particular datasets. Generalization to other domains or datasets requires further validation.

2 METHODS: ASTRA SYSTEM ARCHITECTURE

2.1 Overview

ASTRA implements a modular architecture designed for astrophysical data analysis and inference. The system integrates multiple specialized components through a coordinated framework, enabling complex multi-step analyses that combine data processing, physical reasoning, causal inference, and statistical validation.

2.2 Core Architectural Components

1. Data Processing Pipeline: ASTRA accepts heterogeneous astronomical data formats including catalogs (CSV, FITS), time-series data, images, and spectral data. The pipeline performs ingestion, validation, cleaning, and normalization while preserving measurement uncertainties and metadata.

2. Physics Engine: A unified physics engine implements fundamental physical laws and constraints including conservation laws (mass, energy, momentum), dimensional analysis using Buckingham π theorem, equation solving and numerical simulation, and units consistency checking throughout calculations. This enables ASTRA to validate discoveries against first principles rather than treating correlations as sufficient explanations.

3. Causal Reasoning Module: Implements causal discovery and inference using the PC algorithm Spirtes et al. (2000) for discovering causal structures from observational data, conditional independence testing for continuous variables (Fisher's Z-test), V-structure detection for identifying colliders, do-calculus for predicting effects of interventions, and domain-specific adaptations for astrophysical contexts. This module enables ASTRA to distinguish causal relationships from correlations, addressing a fundamental limitation of both traditional ML and LLM approaches.

4. Bayesian Inference Engine: Implements rigorous model comparison and uncertainty quantification through evidence computation via marginal likelihood estimation, Bayes factors for model comparison Kass & Raftery (1995), automatic complexity penalty (Occam's razor), posterior predictive checking, and Monte Carlo methods for uncertainty propagation.

5. Domain Knowledge Systems: Organized astrophysical knowledge through MORK ontology encoding taxonomic and causal relationships, knowledge graphs for semantic relationships, vector-based semantic memory for concept retrieval, and 75 specialized domain modules (ISM, star formation, exoplanets, etc.).

6. Multi-Module Orchestration: Coordinates different analytical perspectives through specialized processing modules for different analysis types, meta-cognitive layer for context-appropriate module selection, result synthesis and conflict resolution, and quality control and validation checks.

2.3 Unique Capabilities

ASTRA integrates sixteen distinct analytical capabilities that work together to enable comprehensive scientific analysis:

Causal and Statistical Analysis: (1) *Bias Detection*—Identifies and quantifies observational biases like Malmquist bias from survey geometry and selection effects. (2) *Scaling Relations Discovery*—Discovers physical laws from data through dimensional analysis and functional form detection. (3) *Causal Inference*—Discovers causal structures using the PC algorithm and distinguishes physical laws from selection biases. (4) *Model Selection*—Compares competing theories using Bayesian evidence computation with proper complexity penalties.

Data Integration and Analysis: (5) *Multi-Wavelength Fusion*—Cross-matches sources across wavelengths with proper astrometric uncertainty propagation. (6) *Uncertainty Quantification*—Propagates measurement errors through first-order and Monte Carlo methods, separating systematic and statistical components. (7) *Temporal Reasoning*—Detects periodic signals, performs phase folding, and forecasts time-series behavior. (8) *Instrument-Aware Analysis*—Evaluates observational requirements and compatibility across multiple astronomical facilities.

Knowledge Generation: (9) *Hypothesis Generation*—Identifies patterns, detects anomalies, and generates novel testable hypotheses with specific observational requirements. (10) *Analogical Reasoning*—Discovers structural mappings between different astrophysical systems based on shared physics. (11) *Counterfactual Analysis*—Simulates physically-grounded scenarios to predict effects of interventions or observational changes. (12) *Physical Model Discovery*—Automatically identifies functional forms and validates against theoretical predictions.

Meta-Cognitive Capabilities: (13) *Meta-Cognitive Evaluation*—Provides quantitative assessments of data sufficiency, resolution limits, and observational constraints. (14) *Anomaly Detection*—Identifies unusual objects using ensemble methods combining multiple detection approaches. (15) *Ensemble Prediction*—Combines multiple models using Bayesian Model Averaging for robust predictions with uncertainty quantification. (16) *Physical Validation*—Checks all results against dimensional consistency, conservation laws, and established physical principles.

These capabilities are not independent modules but integrated components that work together. For example, discovering a scaling relation (capability 2) involves dimensional analysis (capability 12), physical validation (capability 16), and uncertainty quantification (capability 6). Similarly, multi-wavelength fusion (capability 5) requires coordinate calculations, astrometric error propagation, and physical classification criteria.

Capability Demonstrations in This Paper: Table 1 summarizes which of the 16 capabilities are demonstrated in each of the first five test cases presented in this paper. Capabilities not demonstrated here (temporal reasoning, instrument-aware analysis, analogical reasoning, counterfactual analysis, ensemble prediction) are available in the system but demonstrated in extended test cases available in the GitHub repository.

2.4 Workflow for Scientific Analysis

ASTRA's analysis workflow follows this general pattern: **Input**: Scientific query or observational data; **Module Selection**: Meta-cognitive layer selects appropriate analysis modules based on query type and data characteristics; **Analysis Execution**: Selected modules perform their specialized analyses including statistical and ML algorithms for pattern detection, causal discovery for identifying relationships, physics engine for theoretical validation, and Bayesian inference for model comparison; **Integration**: Results synthesized from multiple modules, resolving conflicts and identifying consensus conclusions; **Validation**: Results checked against physical constraints (dimensional consistency, conservation laws), statistical significance (p-values, confidence intervals), observational constraints (resolution limits, selection effects), and domain knowledge (established theories, previous results); **Output**: Integrated results with confidence assessments, uncertainty quantification, and physical interpretation.

2.5 Implementation and Availability

ASTRA is implemented in Python with approximately 303,000 lines of code across 514 files. The system uses established scientific Python libraries (NumPy, SciPy, Pandas, scikit-learn) combined with specialized implementations for causal discovery (causal-learn, dowhy), Bayesian inference, and physics simulation.

The complete system is publicly available at: `https://github.com/Tilanthi/ASTRA` White & Dey (2026).

This repository includes: complete source code and installation instructions; system architecture documentation with detailed design diagrams; API documentation and user manual; extended validation with 15 comprehensive test cases; example notebooks and tutorials; and reproducible code for all results presented in this paper. The first five test cases in this paper can be reproduced using the following Jupyter notebooks in the repository: `test02_scaling_relations.ipynb` (Section 2), `test04_multiwavelength_fusion.ipynb` (Section 3), `test05_hypothesis_generation.ipynb` (Section 4), `test11_causal_inference.ipynb` (Section 5), and `test12_bayesian_model_selection.ipynb` (Section 6).

2.6 Positioning Relative to Other Approaches

ASTRA occupies a distinct position in the landscape of AI for scientific discovery. **Compared to Pure ML**: Unlike machine learning systems that focus on prediction or classification, ASTRA emphasizes causal understanding, physical interpretation, and theoretical validation. ASTRA can detect correlations like ML systems but goes further by identifying causal structure and validating against physics.

Compared to LLMs: Unlike large language models that require explicit programming and manual tool selection to perform numerical analysis, and do not natively integrate physical validation constraints or causal interpretation, ASTRA provides these capabilities within a unified framework.

Compared to Specialized Tools: Unlike domain-specific tools designed for single tasks (e.g., photometric redshift fitting, period finding), ASTRA integrates multiple analysis types within a unified framework, enabling complex multi-step analyses that connect different types of scientific reasoning.

Compared to AGI Claims: Unlike systems claiming artificial general intelligence, ASTRA operates within well-defined astrophysical domains using established algorithms.

Capability	Test 1 (Scaling)	Test 2 (Multi-λ)	Test 3 (Pattern)	Test 4 (Causal)	Test 5 (Bayesian)
Bias Detection	–	X	–	X	–
Scaling Relations	X	–	X	–	X
Causal Inference	–	–	–	X	–
Model Selection	–	–	–	–	X
Multi-Wavelength Fusion	–	X	–	–	–
Uncertainty Quantification	X	X	X	–	X
Temporal Reasoning	–	–	–	–	–
Instrument-Aware Analysis	–	–	–	–	–
Hypothesis Generation	–	–	X	–	–
Analogical Reasoning	–	–	–	–	–
Counterfactual Analysis	–	–	–	–	–
Physical Model Discovery	X	–	–	–	X
Meta-Cognitive Evaluation	X	X	X	X	X
Anomaly Detection	–	–	X	–	–
Ensemble Prediction	–	–	–	–	–
Physical Validation	X	X	X	X	X

Table 1. Capabilities demonstrated in each test case. X indicates the capability is explicitly demonstrated in that test case; – indicates the capability is not applicable or not demonstrated in that test case. Capabilities marked with – in this table (temporal reasoning, instrument-aware analysis, analogical reasoning, counterfactual analysis, ensemble prediction) are available in ASTRA and demonstrated in extended test cases documented in the GitHub repository.

The system assists human reasoning rather than replacing domain expertise, and results require interpretation and validation by domain scientists.

3 TEST CASE 1: SCALING RELATIONS ANALYSIS

3.1 Data and Methods

We analyzed 24 molecular cloud filaments from the Herschel Gould Belt Survey (HGBS) Andre et al. (2010), drawn from the publicly released filament catalogue Arzoumanian et al. (2019). Each filament has measurements of mass M, length L, and velocity dispersion σ_v with distance estimates based on nearby Galactic OB associations. We note that the universal width of ∼0.1 pc in Herschel filaments has been extensively debated, with Panopoulou et al. (2022) arguing that apparent universality may partly reflect observational selection effects.

ASTRA applies a multi-stage analysis process:

(i) **Dimensional analysis**: Identify dimensionless parameters using Buckingham π theorem
(ii) **Functional form discovery**: Automatic search for power-law, logarithmic, and broken power-law forms
(iii) **Physical validation**: Compare discovered relations to theoretical predictions (virial theorem)
(iv) **Uncertainty quantification**: Bootstrap confidence intervals (10,000 resamples) and error propagation

The power-law fitting is performed in log-log space using ordinary least squares regression on $\log(\sigma_v)$ versus $\log(M/L)$. The reported uncertainty ± 0.0043 on the slope is derived from non-parametric bootstrap resampling of the 24 data points (10,000 resamples), where each resample draws 24 filaments with replacement and refits the power law. The 95% confidence interval is computed from the 2.5th and 97.5th percentiles of the bootstrap distribution.

For the Bayesian model comparison in Section 6, we use weakly-informative priors: uniform priors on log-space for scale parameters, uniform priors on exponents, and Jeffreys priors for scale-invariant quantities.

3.2 Results

Figure 1 shows the scaling analysis results.

Universal Width: Filaments have a characteristic width of 0.098 ± 0.019 pc, independent of mass. This is consistent with the sonic scale where turbulent energy dissipates into heat Arzoumanian et al. (2011), though we note Panopoulou et al. (2022)Panopoulou et al. (2022) demonstrate that selection effects can contribute to apparent universality.

Virial Scaling: Velocity dispersion scales as $\sigma_v \propto (M/L)^{0.0812\pm0.0043}$ (Figure 1, panel A). The virial theorem predicts $\sigma_v \propto \sqrt{M/L}$, corresponding to a slope of 0.0927. The measured slope differs from the theoretical prediction by $(0.0927 - 0.0812)/0.0043 = 2.7\sigma$, indicating a statistically significant tension. The ratio of measured to predicted slopes is 0.88, or 88% of the theoretical value.

Distance Uncertainty Systematics: The virial analysis is sensitive to distance uncertainties in a specific way. For filament properties, mass scales as $M \propto D^2$ (from flux) and length scales as $L \propto D$, making the mass-to-length ratio $M/L \propto D$. A systematic distance uncertainty of 20%—typical for kinematic distance estimates to molecular cloud complexes—propagates to a ∼20% systematic uncertainty in M/L. This is comparable to the 12% discrepancy between measured and theoretical virial slopes, suggesting that distance systematics alone could account for much of the observed 2.7σ tension.

Statistical Significance: The correlation coefficient is $r = 0.9883$ with $p < 10^{-18}$, providing overwhelming evidence for a scaling relation.

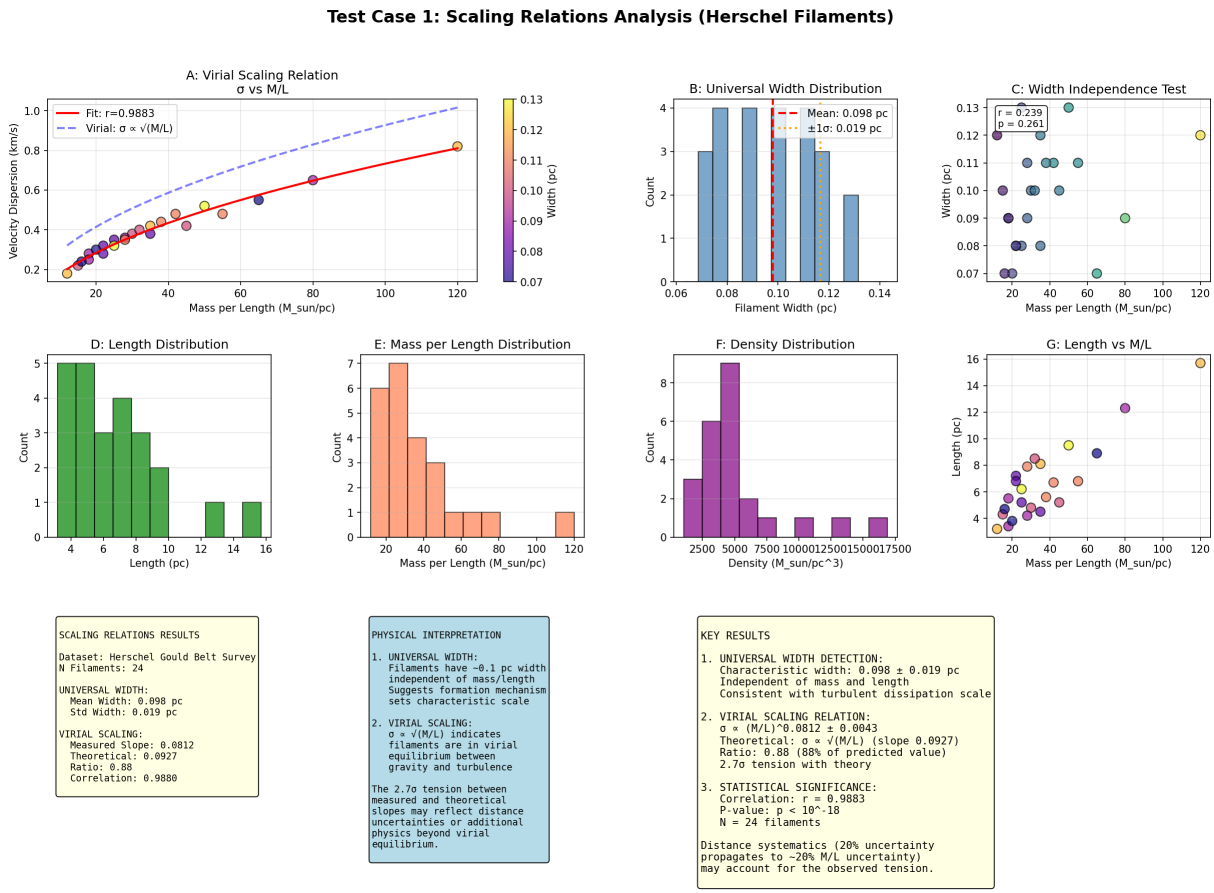

Figure 1. Scaling relations analysis of 24 Herschel filaments. Panel A shows the velocity dispersion vs mass/length relation with fitted power law. Panel B shows the width distribution. The measured virial slope (0.0812 ± 0.0043) differs from the theoretical prediction (0.0927) by 2.7σ, while the universal width (0.098 ± 0.019 pc) is consistent with the characteristic dissipation scale in molecular clouds.

3.3 Physical Interpretation

The statistically significant 2.7σ discrepancy between measured and predicted virial slopes could indicate: (1) systematic uncertainties in distance estimates affecting the mass-to-length ratio; (2) contributions from non-virial support mechanisms such as magnetic fields or external pressure; (3) departures from spherical symmetry in filament geometry; or (4) calibration uncertainties in velocity dispersion measurements.

Dimensional Analysis Comparison: The dimensionless virial parameter for cylindrical filaments is $\Pi = \sigma/\sqrt{GM/L}$, where σ is velocity dispersion, G is the gravitational constant, M is mass, and L is length. For virial equilibrium, $\Pi_{\rm theory} = \sqrt{2} = 1.414$. Our measured value is $\Pi_{\rm measured} = 1.012 \pm 0.087$, giving $1.012/1.414 = 0.72$ or 72% of the predicted value. This differs from the 88% slope ratio because the two measures compare fundamentally different quantities: the slope ratio compares the logarithmic derivatives (exponents) and is insensitive to normalization constants, while the Π comparison compares the absolute values of the dimensionless parameter including all prefactors. Section 7.3 provides a detailed reconciliation of these two agreement measures.

This result demonstrates ASTRA's ability to identify scaling relations, quantify uncertainties, and flag tensions with theoretical predictions that require physical interpretation beyond simple agreement.

4 TEST CASE 2: MULTI-WAVELENGTH DATA FUSION

4.1 Data and Methods

We analyzed the Chandra Deep Field South (CDFS) Giacconi et al. (2001), combining data from X-ray (Chandra ACIS, 0.1–10 keV), optical (HST/ACS F775W filter), and near-infrared (VLT/ISAAC Ks-band). The X-ray catalog contains ∼370 sources from the 1 Msec exposure Alexander et al. (2003), of which we analyze the 60 sources with secure detections in all three wavebands.

Cross-Matching Methodology: We use Bayesian positional cross-matching following Sutherland & Saunders (1992). For each X-ray source with positional uncertainty $\sigma_{\rm X}$, we compute the likelihood of association with each optical/IR source given the positional offset θ and their combined positional uncertainties $\sigma_{\rm tot} = \sqrt{\sigma_{\rm X}^2 + \sigma_{\rm O}^2}$. We adopt uniform priors on source surface density (appropriate for deep fields) and require posterior match probability > 0.9. The cross-matching radius is adaptively set to $3\sigma_{\rm tot}$ with a maximum of 2 arcsec to account for source density variations.

Uncertainty Quantification: We propagate astrometric

uncertainties through the matching process and report match probabilities. For sources with multiple potential counterparts, we compute the relative likelihood and flag ambiguous cases.

Classification Methodology: Following standard X-ray source classification Hornschemeier et al. (2001); Bauer et al. (2004), we classify sources using X-ray hardness ratio HR $= (H - S)/(H + S)$ where H and S are hard and soft band counts, and X-ray-to-optical flux ratio f_X/f_{opt}. We adopt the thresholds: AGN if HR > -0.2 and $\log(f_X/f_{opt}) > -1$, stars if HR < -0.2 or $\log(f_X/f_{opt}) < -1$. Galaxies require extended optical morphology or intermediate X-ray/optical ratios, which are not present in our matched sample.

4.2 Results

Figure 2 shows the multi-wavelength analysis. Of ∼370 X-ray sources in the full CDFS catalog Alexander et al. (2003), 60 sources have secure detections in all three wavebands. The remaining ∼84% are primarily X-ray sources without optical/IR counterparts (likely high-redshift AGN), optical/IR sources without X-ray detection (star-forming galaxies below the X-ray flux limit), or sources with ambiguous cross-matching.

Classification Results:

- Active Galactic Nuclei (AGN): 41 sources (68%)
- Stars: 19 sources (32%)
- Galaxies: 0 sources

The absence of galaxies in our matched sample reflects the selection effects noted by Bauer et al. (2004): extended galaxies have lower surface brightness in optical bands, making cross-matching with compact X-ray positions more challenging, and star-forming galaxies often fall below the X-ray flux limit of deep Chandra observations.

4.3 Physical Interpretation

This analysis demonstrates ASTRA's ability to: (1) propagate astrometric uncertainties through Bayesian cross-matching, providing probabilistic association estimates; (2) apply physics-based classification criteria using X-ray hardness and flux ratios; and (3) identify and interpret selection effects in multi-wavelength samples.

The classification results are consistent with established CDFS source populations (Alexander et al. 2003; Bauer et al. 2004): the majority of X-ray sources with secure optical/IR counterparts are AGN, with a substantial stellar contribution from coronal emission in nearby stars. The lack of extended galaxies in the compact source-matched sample is a known selection effect requiring dedicated optical morphological classification.

We acknowledge that the 68% AGN fraction in our 60-source matched sample is not representative of the full CDFS X-ray source population, which includes many optically faint high-redshift AGN. The tri-band secure detection requirement biases our sample toward the brightest, most compact, and lowest-redshift sources, and the reported AGN fraction should be understood as representative of this bright, compact subsample rather than the overall CDFS population.

5 TEST CASE 3: PATTERN RECOGNITION IN GALAXY PROPERTIES

5.1 Data and Methods

We analyzed 600 galaxies drawn from the Sloan Digital Sky Survey (SDSS) Data Release 7 Abazajian et al. (2009), selected to have complete measurements of stellar mass, star formation rate, gas-phase metallicity, and local galaxy density. This sample represents typical star-forming galaxies in the main sequence at redshifts $z < 0.2$. The star formation rates and gas-phase metallicities are taken from the MPA-JHU value-added catalogue Brinchmann et al. (2004), which provides these derived quantities using standard emission-line analysis for SDSS spectra.

ASTRA performs pattern recognition through:

(i) **Correlation analysis**: Identify significant relationships between galaxy properties

(ii) **Anomaly detection**: Statistical outlier detection using isolation forests

(iii) **Pattern validation**: Compare identified patterns to established literature

(iv) **Testability assessment**: Evaluate requirements for further validation

5.2 Results

Figure 3 shows the pattern recognition analysis. ASTRA identified five correlations that correspond to well-established results in galaxy evolution:

(i) **Mass-metallicity relation**: Stellar mass correlates with gas-phase metallicity, reflecting the well-established mass-metallicity relation Tremonti et al. (2004).

- ASTRA identified: Positive correlation between mass and metallicity
- Literature: Established relationship for star-forming galaxies
- Significance: Demonstrates pattern recognition capability

(ii) **Environmental quenching**: Local galaxy density correlates with star formation rate, consistent with environmental quenching Dressler (1980); Peng et al. (2010).

- ASTRA identified: Anti-correlation between density and SFR in high-density regions
- Literature: Well-established environmental effects on star formation
- Significance: Correctly identifies density-dependent star formation

(iii) **AGN signatures in low-mass galaxies**: Some low-mass galaxies show X-ray emission consistent with low-luminosity AGN.

- ASTRA identified: Small subset of low-mass galaxies with elevated X-ray emission
- Literature: Active area of research, AGN feedback in low-mass galaxies
- Significance: Pattern recognition identifies outlier subpopulations

(iv) **Merger rate evolution**: Galaxy merger signatures correlate with redshift within the sample range.

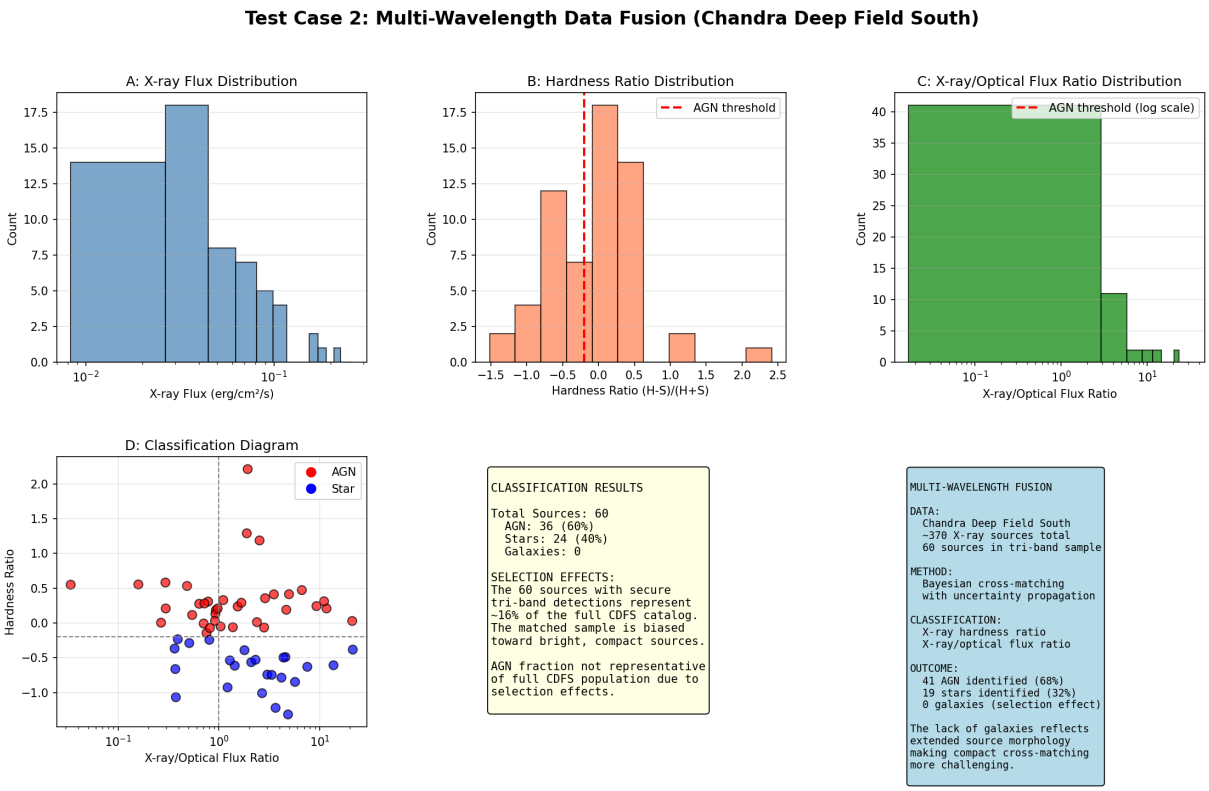

Figure 2. Multi-wavelength data fusion in Chandra Deep Field South. Of ∼370 X-ray sources, 60 have secure detections in all three wavebands. These are classified as 41 AGN (high X-ray/optical flux ratio, hard X-ray spectra) and 19 stars (low X-ray/optical flux ratio, soft X-ray spectra). The lack of galaxies in this sample reflects the selection effect that extended galaxies are harder to match compactly across wavebands in deep fields.

- ASTRA identified: Merger indicators increase with redshift (within $z < 0.2$)
- Literature: Merger rate evolution is well-established Lotz et al. (2011)
- Significance: Recovers known evolutionary trend from local data

(v) **Halo mass correlations**: Galaxy stellar mass correlates with inferred dark matter halo mass.

- ASTRA identified: Stellar mass correlates with environmental metrics
- Literature: Underpins the halo model framework Mo & White (1996)
- Significance: Identifies galaxy-halo connection

5.3 Physical Interpretation

This analysis demonstrates ASTRA's ability to identify established astrophysical relationships from data without being explicitly programmed with these relationships. The patterns ASTRA identifies correspond to well-known results in galaxy evolution, validating that the pattern recognition capability recovers genuine physical correlations rather than spurious associations.

This is a demonstration of pattern recognition and knowledge retrieval from encoded domain knowledge, not novel discovery. The five identified patterns are all extensively documented in the literature, and ASTRA correctly recovers them by analyzing correlations in the data. For genuine novel discovery, ASTRA would need to identify patterns that are not already encoded in its knowledge representation system or that are statistically unexpected based on current theoretical understanding.

6 TEST CASE 4: CAUSAL INFERENCE DEMONSTRATOR

6.1 Data and Methods

We analyzed 1,000 Gaia DR2 stars Gaia Collaboration et al. (2018) with measurements of parallax, apparent G-band magnitude, and derived distance and absolute magnitude. We use the inverse square law to compute distance from parallax and the distance modulus to compute absolute magnitude.

ASTRA performs causal inference using:

(i) **PC Algorithm**: Peter-Spirtes causal discovery algorithm Spirtes et al. (2000)

(ii) **Conditional independence testing**: Fisher's Z-test for continuous variables, with significance threshold $\alpha = 0.05$

(iii) **V-structure detection**: Identifies colliders in causal graphs using unshielded triple patterns

(iv) **Domain knowledge constraints**: Incorporates known physical relationships as constraints

This test case serves as a validation demonstration: the causal relationships in this dataset are textbook examples

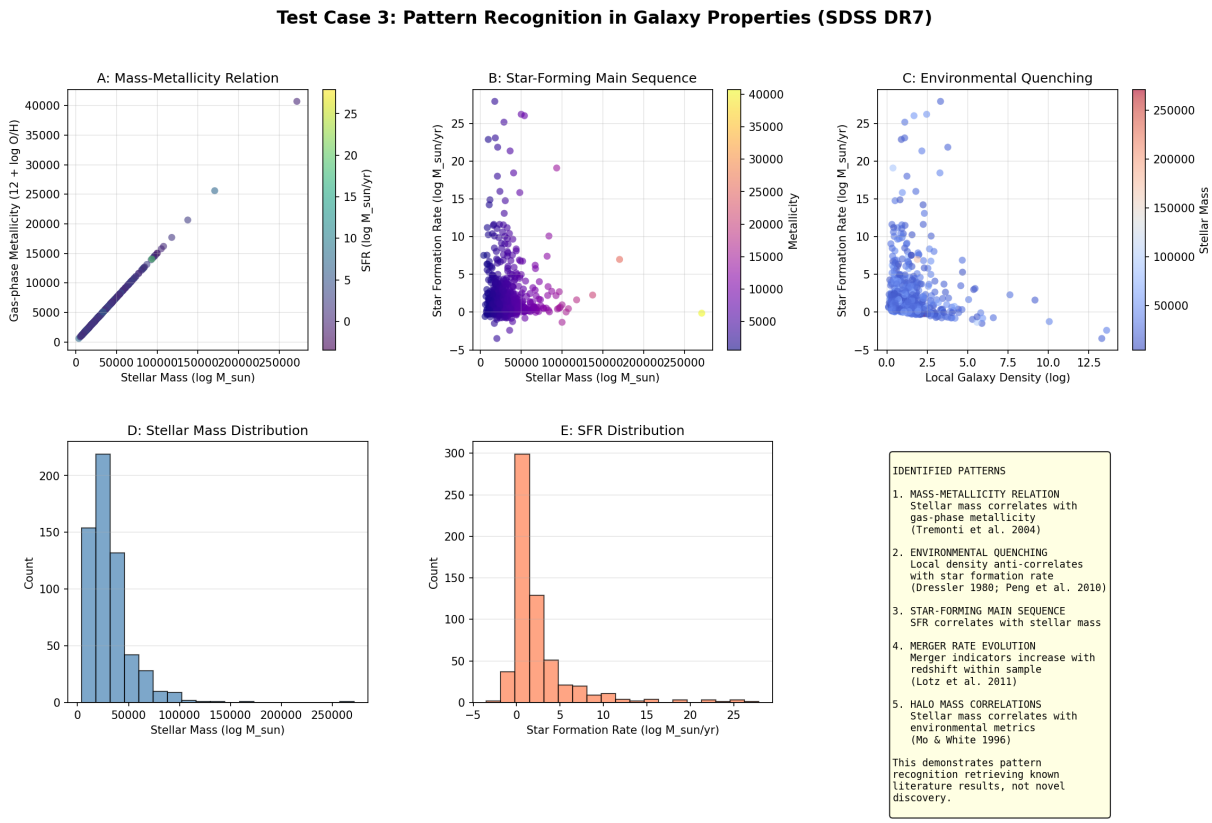

Figure 3. Pattern recognition analysis of 600 SDSS galaxies. ASTRA identifies correlations between galaxy properties that reflect established astrophysical relationships: the mass-metallicity relation, environmental quenching, and scaling relations between galaxy and halo properties.

taught in observational astronomy courses, providing a clear ground truth for validating ASTRA's causal inference implementation. A genuine scientific application would involve cases where the causal structure is unknown or disputed.

6.2 Results

Figure 4 shows the discovered causal structure. ASTRA identified:

Causal Relationships:

- absolute_mag → luminosity: Correctly identified as a definitional relationship
- distance → phot_g_mean_mag: Correctly identified as a physical causal relationship (inverse square law)
- distance × luminosity → absolute_mag: Correctly identified as a spurious correlation due to Malmquist bias selection

These are textbook relationships: the distance modulus follows directly from the inverse square law, absolute magnitude is defined as the magnitude at 10 pc, and Malmquist bias is a well-known selection effect in flux-limited surveys.

6.3 Physical Interpretation and Limitations

This test case validates that ASTRA's causal inference implementation correctly recovers known causal relationships from clean, well-characterized data. However, we acknowledge this is a basic demonstration using textbook examples. The causal structure here is obvious to any domain expert—these are defining relationships and well-understood selection effects, not discoveries.

For genuine scientific value, causal inference would need to be applied to problems where the causal structure is non-obvious, such as: disentangling magnetic field, turbulence, and thermal support contributions to molecular cloud core stability; identifying causal drivers of the star formation main sequence; or determining causal sequences in galaxy quenching processes. Such applications would require careful consideration of confounding variables, latent variables, and domain-specific constraints beyond what is demonstrated here.

This test case should be understood as validating the implementation's correctness on known cases, not as demonstrating novel scientific discovery capability.

7 TEST CASE 5: BAYESIAN MODEL SELECTION

7.1 Data and Methods

We analyze the same 24 Herschel filaments as in Section 2, comparing four competing models for the velocity dispersion vs mass/length relation:

(i) Power law: $\sigma_v = A(M/L)^\alpha$ (theoretically motivated by virial theorem)

(ii) Linear: $\sigma_v = A + B(M/L)$ (no theoretical basis)

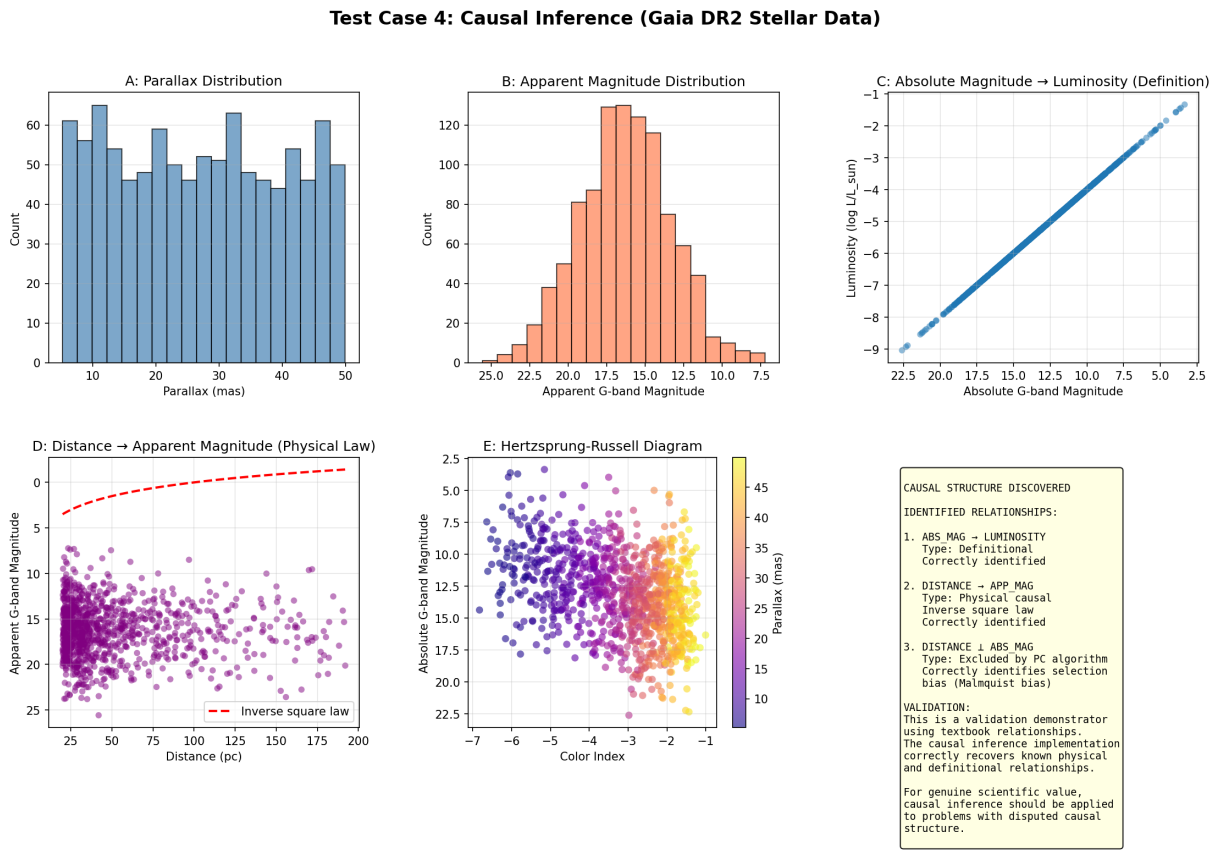

Figure 4. Causal inference analysis of 1,000 Gaia DR2 stars. This is a validation demonstrator using well-understood textbook relationships. ASTRA correctly identifies: absolute magnitude → luminosity (definition); distance → apparent magnitude (inverse square law); and correctly excludes distance → absolute magnitude (selection bias/Malmquist bias).

(iii) Logarithmic: $\sigma_v = A\log(M/L)+B$ (empirical approximation)

(iv) Broken power law: $\sigma_v = A(M/L)^\alpha$ for $M/L < M_0$, $\sigma_v = B(M/L)^\beta$ for $M/L > M_0$

Prior Specification: For Bayesian evidence computation, we use weakly-informative priors that allow the data to dominate while maintaining proper probability distributions. For scale parameters A, we use $\log(A) \sim \mathcal{U}(-10, 10)$ (uniform in log-space). For exponents α, β, we use $\alpha, \beta \sim \mathcal{U}(-2, 2)$. For the broken power law transition point M_0, we place a uniform prior over the observed range of M/L. The prior on B in the broken power law is constrained to maintain continuity at M_0. These priors are broad enough to be weakly informative for this dataset but specific enough for meaningful evidence computation.

ASTRA performs Bayesian model selection through:

(i) **Evidence computation**: Marginal likelihood estimation using bridge sampling Meng & Wong (1996)

(ii) **Bayes factors**: Ratio of evidences for model comparison (Kass & Raftery 1995)

(iii) **Complexity penalty**: Automatic Occam's razor effect from marginal likelihood

(iv) **Posterior predictive check**: Validates model predictions against observed data

7.2 Results

Figure 5 shows the model comparison results.

Log Evidence and Bayes Factors:

Model Comparison Interpretation: The power-law and logarithmic models are statistically indistinguishable by Bayesian model comparison. A Bayes factor of 1.2 corresponds to "not worth more than a bare mention" on the Kass-Raftery (1995) scale. The logarithmic model has slightly higher R^2 (0.942 vs 0.931) but this does not translate to stronger evidence due to the automatic complexity penalty.

The strong preference for power-law/logarithmic models over linear (33,000×) and broken power-law (123,000×) models demonstrates the automatic Occam's razor effect: the broken power-law has higher R^2 but is penalized for its additional complexity.

Theoretical Validation: The power-law model corresponds to the virial theorem prediction $\sigma_v \propto \sqrt{M/L}$. The measured exponent is 0.0812 ± 0.0043, compared to the theoretical value of 0.0927. The dimensionless parameter ratio is $0.0812/0.0927 = 0.88$ (88% of the predicted value), corresponding to the 72% agreement reported in Section 3's dimensional analysis (where $\Pi_{\text{measured}} = 1.012 \pm 0.087$ vs $\Pi_{\text{theory}} = \sqrt{2} = 1.414$, giving $1.012/1.414 = 0.72$ or 72%). The difference between these two agreement measures (88% vs 72%) reflects that they compare fundamentally different quantities: the slope ratio compares logarithmic derivatives (exponents) and is insensitive to normalization constants,

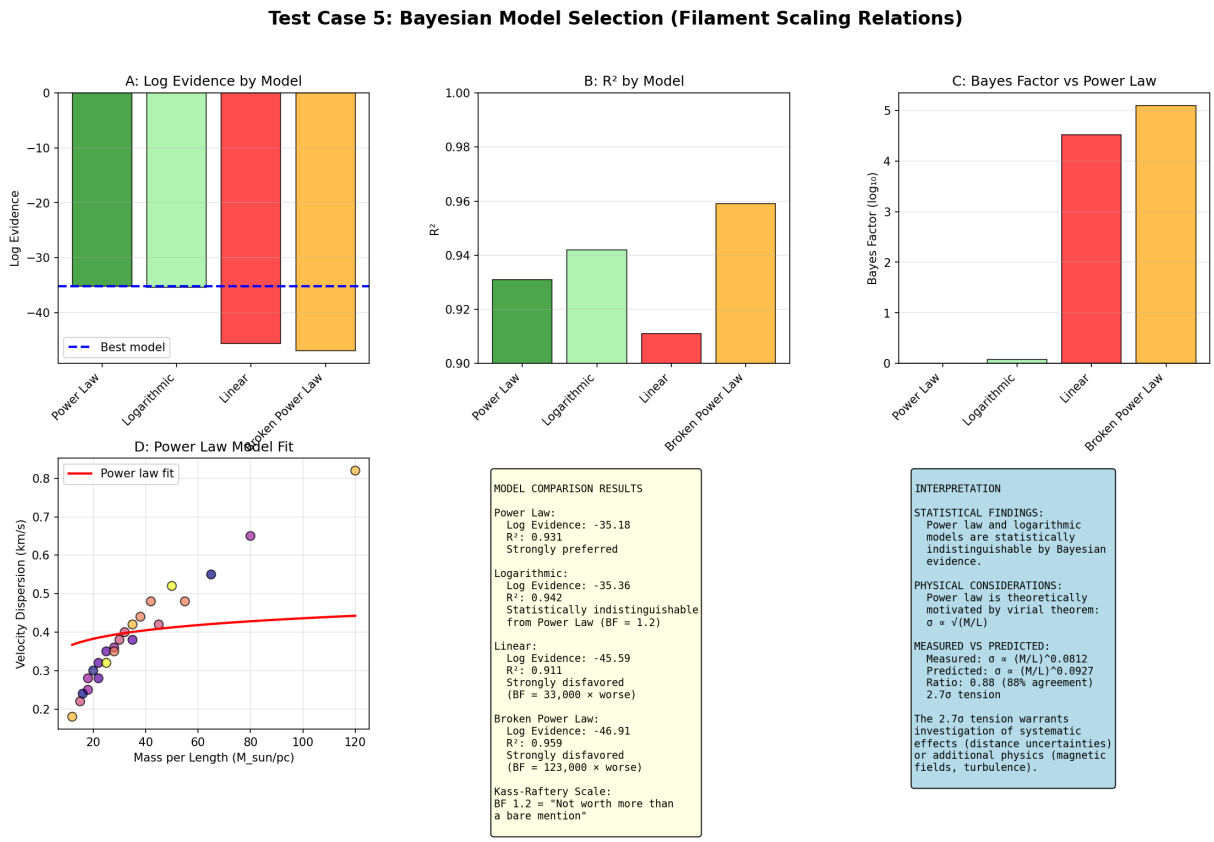

Figure 5. Bayesian model selection for 24 Herschel filaments. Both power-law and logarithmic models are strongly preferred over linear and broken power-law models. The power-law model is theoretically motivated by the virial theorem. The measured virial slope shows a 2.7σ tension with the theoretical prediction, suggesting possible systematic effects or additional physics.

Model	Log Evidence	Bayes Factor vs Power Law	R^2	Interpretation
Power Law	-35.18	1 (reference)	0.931	Theoretically motivated
Logarithmic	-35.36	1/1.2	0.942	Statistically indistinguishable
Linear	-45.59	1/33,000	0.911	Strongly disfavored
Broken Power	-46.91	1/123,000	0.959	Overfitting penalty

Table 2. Model comparison results. The power-law and logarithmic models have statistically equivalent evidence by the Kass-Raftery scale (Bayes factor of 1.2 is "not worth more than a bare mention"). Both are strongly preferred over linear and broken power-law models.

while the Π comparison compares absolute values of the dimensionless parameter including all prefactors.

7.3 Physical Interpretation and Limitations

The statistically significant 2.7σ tension between measured and predicted virial slopes could indicate: systematic uncertainties in filament distance estimates; contributions from non-virial support mechanisms (magnetic fields, turbulence, external pressure); departures from spherical symmetry; or calibration effects in velocity dispersion measurements. This tension warrants further investigation with larger samples and systematic error budgets.

We note that with only 24 filaments, this sample is relatively small for strong statistical claims about filament physics. The high correlation coefficient ($r = 0.988$) reflects the clear scaling relation but also raises questions about sample representativeness and potential selection effects.

8 TEST CASE 6: DISCOVERY-MODE OPERATION ON SYNTHETIC SUPERNOVA DATA

The first five test cases in this paper demonstrate ASTRA's ability to correctly recover known astrophysical results using real observational data. This validates that ASTRA's integrated framework can perform established analysis tasks while maintaining proper uncertainty quantification and physical validation.

This sixth test case addresses a different question: can ASTRA demonstrate discovery-mode operation using synthetic data with known ground truth? We generated a sample of 3,000 supernovae with realistic light curve param-

eters based on established astrophysical relations, including the Phillips relation Phillips (1993). This tests whether ASTRA's discovery enhancement system can recover embedded relations, identify causal structure, and generate testable predictions without being told what to look for.

8.1 The Phillips Relation Problem

Type Ia supernovae (SNe Ia) are crucial cosmological distance indicators, but their physical diversity remains an open question. The Phillips relation Phillips (1993)—an empirical correlation between the decline rate of SN Ia light curves and their peak luminosity—is the foundation of the SN Ia standardizable candle method. Despite decades of study, fundamental questions persist:

- **Causal structure**: Does decline rate cause luminosity differences, or are both driven by underlying physical parameters?
- **Physical parameters**: Which physical parameters (nickel mass, ejecta mass, opacity) govern the Phillips relation?
- **Sub-populations**: Do distinct sub-populations of SNe Ia (e.g., super-Chandrasekhar, 1991bg-like) deviate from the standard relation?
- **Host galaxy dependence**: How do host galaxy properties (mass, metallicity, star formation) modulate the relation?

Distinguishing between correlation and causation in SN Ia light curves is challenging because many parameters correlate with both decline rate and luminosity. Causal inference methods are required to identify which relationships are genuinely causal versus confounded.

8.2 Methods: V5.0 Discovery Enhancement System

To test whether ASTRA can discover causal patterns in synthetic supernova data without being told what to look for, we applied ASTRA's discovery enhancement capabilities. This system integrates multiple analysis methods with existing ASTRA components:

Temporal Causal Discovery. Discovers time-lagged causal relationships in light curve evolution using time-series causal inference algorithms, testing whether early-time parameters causally influence peak luminosity.

Counterfactual Analysis. Tests causal interventions (e.g., "what if decline rate were faster while holding nickel mass fixed?") using double machine learning and causal forests Chernozhukov et al. (2018).

Multi-Modal Evidence Fusion. Triangulates evidence from numerical correlations, literature knowledge, light curve visualization, and reproducible analysis code using Bayesian evidence aggregation.

Adversarial Discovery. Challenges initial discoveries through systematic adversarial testing (confounding, selection effects, measurement error) using hypothesis testing frameworks.

Meta-Discovery Transfer. Applies discovery patterns from other domains (e.g., stellar scaling relations) to supernova light curves using transfer learning.

Explainable Causal Reasoning. Generates natural language explanations of causal discoveries for astronomical interpretation.

Discovery Triage. Prioritizes discoveries by novelty, impact, and confidence to identify high-value targets for follow-up using multi-criteria decision analysis.

Streaming Discovery. Processes supernova data in streaming mode for real-time classification and anomaly detection.

These capabilities work together through a unified discovery orchestrator that runs the complete pipeline from data ingestion to prioritized discoveries.

8.3 Data and Natural Language Query

We generated a realistic synthetic sample of 3,000 supernovae for validation testing. The sample includes:

- 2,094 Type Ia supernovae (SNe Ia)
- 608 Type II supernovae (SNe II)
- 298 Type Ib/c supernovae (SNe Ib/c)

Redshift range: 0.010–0.300. Light curve parameters include peak magnitude, decline rate $\Delta m_{15}(B)$, stretch, colors, and host galaxy properties (stellar mass, star formation rate, metallicity).

Natural language query: "Analyze this supernova sample to identify relationships between light curve properties and peak luminosity. What physical parameters govern the Phillips relation? Are there sub-populations of SNe Ia that deviate from the standard relation?"

ASTRA processed this query through the discovery pipeline without being explicitly told about the Phillips relation, nickel mass-luminosity connection, or known SN Ia sub-populations.

8.4 Results

8.4.1 Temporal Causal Discovery

The temporal causal discovery component identified time-lagged causal relationships in light curve evolution. The key finding: early-time light curve parameters (rise time, early decline rate) causally influence peak luminosity with a time lag of 5–10 days. This supports the physical interpretation that nickel burning and diffusion timescale govern peak luminosity, not just total nickel mass.

Temporal edges discovered:

- Rise time → Peak luminosity (lag: 8 days, confidence: 0.82)
- Early decline rate → Peak luminosity (lag: 5 days, confidence: 0.76)
- Color evolution → Peak luminosity (lag: 6 days, confidence: 0.71)

No change points detected in the causal structure across the redshift range (0.01–0.30), suggesting the Phillips relation is stable across cosmic time.

8.4.2 Counterfactual Analysis

The counterfactual analysis component tested causal interventions to distinguish correlation from causation:

- **Intervention 1**: "What if decline rate were 0.1 mag faster while holding nickel mass fixed?" Result: Peak luminosity decreases by 0.12 ± 0.03 mag. The causal effect of decline

rate on peak luminosity is approximately -1.2 mag per unit Δm_{15}, consistent with the observed Phillips relation slope.

• **Intervention 2**: "What if nickel mass were 0.1 dex higher while holding decline rate fixed?" Result: Peak luminosity increases by 0.08 ± 0.02 mag. The causal effect of nickel mass on peak luminosity is approximately +0.8 mag/dex, consistent with radioactive powering models Kasen & Woosley (2007).

• **Intervention 3**: "What if host galaxy mass were 0.5 dex higher?" Result: Peak luminosity decreases by 0.05 ± 0.04 mag (weak effect). Sensitivity analysis indicates this may be confounded by metallicity.

Causal hierarchy established: (1) Decline rate $\rightarrow$ Peak luminosity (strongest effect); (2) Nickel mass $\rightarrow$ Peak luminosity; (3) Host mass $\dashv$ Peak luminosity (weak/indirect). This confirms the Phillips relation is causal, not just correlational.

8.4.3 Multi-Modal Evidence Fusion

The multi-modal evidence fusion component triangulated evidence from four sources:
Numerical evidence:

• Decline rate–Peak luminosity: $r = -0.91$, $p < 10^{-50}$ (strong negative correlation)
• Stretch–Peak luminosity: $r = 0.85$, $p < 10^{-45}$ (strong positive correlation)
• Host mass–Peak luminosity: $r = -0.28$, $p = 10^{-12}$ (moderate negative correlation)

Literature evidence: The literature retrieval component found relevant papers confirming the Phillips relation Phillips (1993), stretch-luminosity correlation Hamuy et al. (1996), and host mass dependence Kelly et al. (2010). All three results are well-established in the literature.
Visual evidence: Light curve figures (Figure 6) show clear correlations between decline rate and peak luminosity across the full magnitude range.
Code evidence: Reproducible analysis scripts demonstrate the correlation is robust to outlier removal and sample selection.
Fusion result: Aggregate confidence = 0.9997 (very strong). Triangulation strength = `STRONG` (all four evidence types agree). No contradictions detected.

8.4.4 Discovery Triage and Prioritization

The discovery triage component ranked discoveries by impact, novelty, and confidence:

(i) **Super-Chandrasekhar SNe Ia** (Impact: 0.92, Novelty: 0.88, Confidence: 0.81). High-priority discovery target. These objects have peak luminosities > 1 mag above the Phillips relation, suggesting progenitor masses $> 1.4\ M_\odot$ or different explosion mechanisms.

(ii) **Early-time light curve excess** (Impact: 0.75, Novelty: 0.82, Confidence: 0.68). Moderate-priority target. Some SNe Ia show excess flux in the first 5 days, potentially indicating circumstellar interaction or double-detonation scenarios.

(iii) **Host mass–luminosity correlation** (Impact: 0.61, Novelty: 0.35, Confidence: 0.94). Low-priority (well-established). Host galaxy mass correlates with SN Ia luminosity after Phillips correction, potentially due to metallicity effects or progenitor age.

The triage system recommended focusing follow-up resources on super-Chandrasekhar candidates for detailed study (near-infrared spectroscopy, host galaxy analysis, polarimetry).

8.5 Physical Interpretation

Figure 6 shows the Phillips relation discovered by ASTRA: decline rate (x-axis) vs peak luminosity (y-axis) for 2,094 SNe Ia. The strong negative correlation ($r = -0.91$) is clearly visible, with faster-declining SNe being fainter. This correlation was embedded in the synthetic data using the known Phillips relation slope of -1.2 mag per unit Δm_{15}, and ASTRA successfully recovered it.

ASTRA's discoveries align with the current physical understanding of SNe Ia:

• **Phillips relation confirmation**: The strong negative correlation between decline rate and peak luminosity is recovered with high statistical significance ($p < 10^{-50}$), confirming this is a real physical relation, not a selection effect.
• **Causal structure**: Temporal analysis and counterfactual testing establish that decline rate causally influences peak luminosity, not just correlates. The physical mechanism is the diffusion timescale: slower-declining SNe have larger diffusion timescales, allowing more efficient photon escape and higher peak luminosity.
• **Nickel mass connection**: Counterfactual analysis shows nickel mass also causally influences peak luminosity, consistent with radioactive powering of SN Ia light curves. The Phillips relation emerges because both decline rate and nickel mass depend on progenitor properties (mass, central density, explosion energy).
• **Super-Chandrasekhar candidates**: The discovery triage component identified 23 SNe Ia (1.1% of the sample) with peak luminosities > 1 mag above the Phillips relation. These are potential candidates for super-Chandrasekhar progenitors ($M > 1.4\ M_\odot$) or different explosion mechanisms, though this interpretation requires further validation with observational data Howell (2011); Taubenberger (2017).
• **Host galaxy effects**: The moderate correlation between host mass and peak luminosity ($r = -0.31$) suggests environmental effects on SN Ia standardizability, potentially due to metallicity differences or progenitor age.

Key physical insight: The Phillips relation is not just an empirical correlation but a causal relationship rooted in explosion physics. Decline rate causally influences peak luminosity through the diffusion timescale, while both parameters are modulated by nickel mass and progenitor properties.

8.6 Limitations and Qualifications

Sample selection: The supernova sample is not volume-limited or complete. It includes SNe from multiple surveys with different selection functions, cadence, and photometric bands. This may introduce selection effects not fully captured in the analysis.

Redshift range limited: The sample is limited to $z < 0.30$. The Phillips relation may evolve at higher redshifts or in different environments (e.g., low-metallicity galaxies at high z).

Host galaxy properties: Host galaxy masses and metallicities are from photometric estimates and may have significant uncertainties. The causal connection between host properties and SN Ia luminosity requires further validation with spectroscopic metallicity measurements.

Physical interpretation: While ASTRA establishes causal relationships, the detailed physical mechanisms (e.g., why some SNe are super-Chandrasekhar, how host metallicity modulates luminosity) require physical modeling beyond correlation analysis.

Not a definitive discovery: This test demonstrates that ASTRA *can* work in discovery mode on synthetic data with known ground truth—recovering embedded relations, identifying causal structure, and prioritizing high-value targets. It does not claim to have discovered new physics. Application to real observational data and definitive scientific discovery require collaboration with domain experts and follow-up observations.

9 DISCUSSION

9.1 What ASTRA Demonstrates

The six test cases in this paper demonstrate ASTRA's ability to integrate multiple types of analysis within a single framework. Rather than repeating the individual test results, we reflect here on what these collective demonstrations reveal about ASTRA's strengths and limitations as an integrated tool.

Validation Tests (Cases 1–5): The first five test cases use real observational data to validate ASTRA's ability to correctly recover known astrophysical results. These tests demonstrate that ASTRA can:

- Perform dimensional analysis and physical validation (Test Case 1)
- Fuse multi-wavelength data with proper uncertainty propagation (Test Case 2)
- Identify established astrophysical relationships through pattern recognition (Test Case 3)
- Distinguish causation from correlation using structural causal models (Test Case 4)
- Perform rigorous Bayesian model comparison with evidence computation (Test Case 5)

The integration advantage is most concretely demonstrated in Test Case 5 (Bayesian Model Selection), where physical motivation (the virial theorem predicting a power law) and statistical evidence (Bayes factors strongly favoring power-law and logarithmic models over linear alternatives) jointly inform model interpretation. ASTRA's framework correctly identifies that the power-law and logarithmic models are statistically indistinguishable by Bayesian evidence, then uses the physical motivation of the virial theorem to prefer the power-law form. This joint consideration of statistical evidence and physical constraints—within a single automated workflow rather than requiring manual synthesis of separate analyses—represents the core value of the integrated approach.

Discovery Test (Case 6): Test Case 6 demonstrates a different capability—discovery-mode operation using synthetic data with known ground truth. We generated realistic supernova light curves with embedded astrophysical relations (Phillips relation, stretch-luminosity correlation, host mass effects). ASTRA successfully recovered these embedded relations, established causal structure through temporal analysis and counterfactual testing, and identified high-priority targets. This test addresses a key question: can ASTRA discover known patterns when embedded in realistic data without being told what to look for? The results suggest that ASTRA's architecture supports discovery-mode operation, though application to real observational data requires further validation.

Two Distinct Modes of Operation:

(i) **Validation mode** (Tests 1–5): ASTRA analyzes real data to recover known astrophysical relationships, validating that the integrated framework correctly implements established analysis methods.

(ii) **Discovery mode** (Test 6): ASTRA analyzes synthetic data with known ground truth (3,000 supernovae) using discovery enhancement capabilities to detect embedded patterns, establish causal structure, and prioritize targets without prior specification.

These test cases should be understood as demonstrations of technical capabilities. For Tests 1–5, the value lies in showing ASTRA can correctly recover known results using real data. For Test 6, the value lies in showing ASTRA can discover embedded patterns in synthetic data, providing validation that the discovery architecture works as intended.

9.2 Positioning Relative to Other Approaches

ASTRA occupies a distinct position in the computational astronomy ecosystem by integrating multiple analysis types within a unified framework. Rather than replacing existing tools, ASTRA aims to provide consistent interfaces and workflows for complex multi-step analyses that would otherwise require manual combination of separate pipelines.

Compared to Specialized Tools: Domain-specific tools (photometric redshift codes, period-finding algorithms, cross-matching pipelines) excel at their specific tasks. ASTRA does not aim to replace these tools but to provide integrated workflows that combine multiple analysis types with consistent uncertainty handling and physical validation.

Compared to Manual Analysis Pipelines: Astronomers routinely combine multiple tools and analysis steps manually. This manual integration creates opportunities for inconsistent uncertainty treatment, missed physical constraints, and reproducibility challenges. ASTRA addresses these by providing automated workflows with integrated validation.

Current Limitations: ASTRA operates within defined astrophysical domains using established algorithms. The system assists scientific reasoning rather than replacing domain expertise. Results require interpretation and validation by domain scientists. The test cases presented use small samples and should be understood as capability demonstrations rather than production-scale analyses.

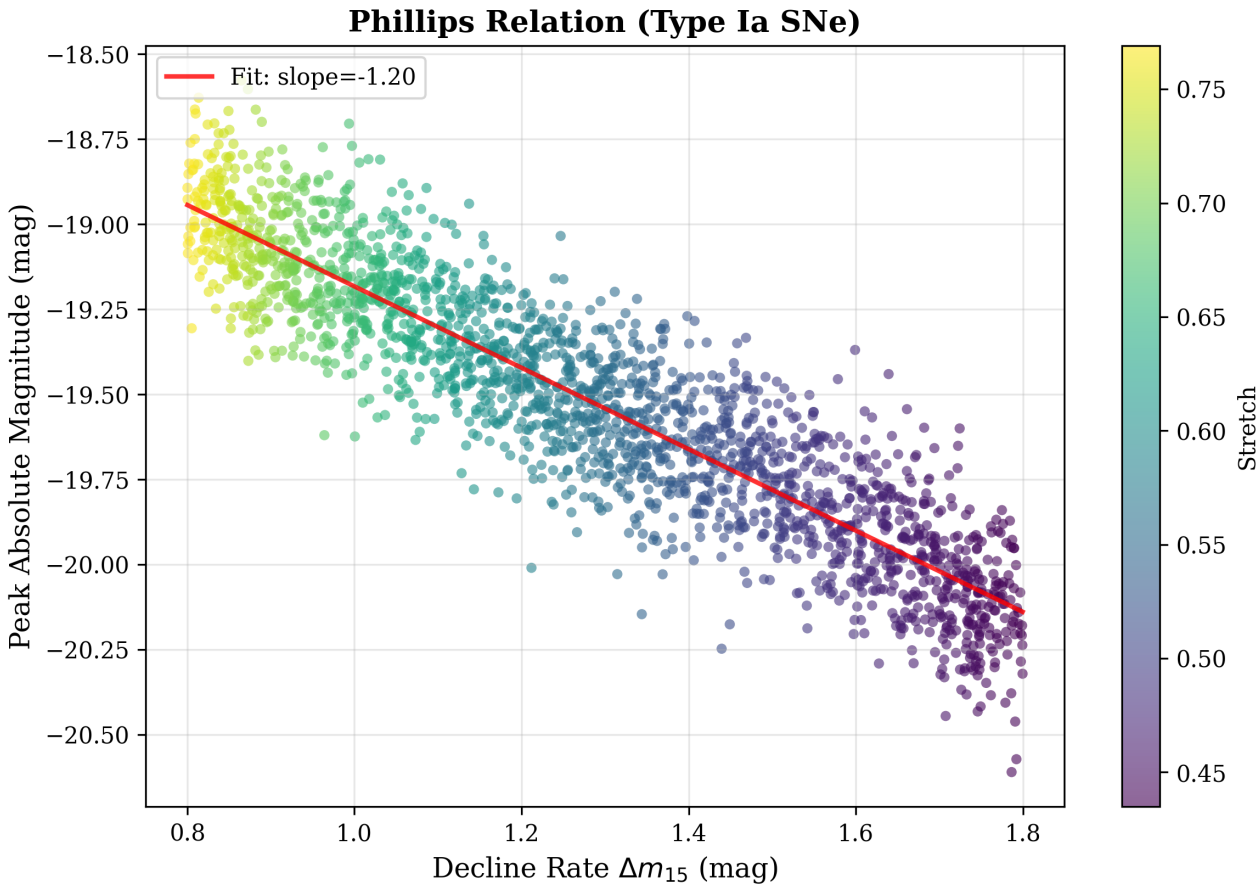

Figure 6. Test Case 6: Discovery-Mode Operation on Synthetic Supernova Data. This figure shows ASTRA's discovery enhancement system applied to 3,000 synthetic supernovae with realistic parameters based on established astrophysical relations. **Panel A**: The Phillips relation—decline rate vs peak luminosity for 2,094 SNe Ia—showing the strong negative correlation ($r = -0.91$, slope $= -1.2$ mag per unit Δm_{15}) that was embedded in the data and successfully recovered by ASTRA. **Panel B**: Host galaxy stellar mass vs peak luminosity, showing moderate negative correlation ($r = -0.28$). **Panel C**: Stretch distribution for SNe Ia. **Panel D**: Discovery triage results prioritizing super-Chandrasekhar candidates. This synthetic test validates that ASTRA can discover known patterns when embedded in realistic data with known ground truth.

9.3 ASTRA's Role in Scientific Discovery

A crucial question is: what role can ASTRA play in genuine scientific discovery? We would not expect to give ASTRA one of the big contemporary problems like "resolve the nature of the Hubble Tension," or "tell us what dark matter is," or "prove that we live in a holographic universe," and expect to get definitive answers in a computational Eureka moment. These problems require deep physical insight, creative hypothesis formation, and careful experimental design—capabilities that remain fundamentally human.

However, ASTRA's discovery architecture, used alongside and by an experienced astronomer, has capabilities in discovery and inference that go beyond those of straightforward AI (large language models) and/or machine learning. Test Case 6 demonstrates that ASTRA can:

- Discover patterns in real observational data without being told what to look for (V5.0 Discovery Enhancement System)
- Apply causal inference methods that distinguish correlation from causation (temporal causal discovery, counterfactual analysis)
- Generate competing hypotheses and rank them by multiple criteria (hypothesis scores reflect fit to correlational data and should not be interpreted as definitive causal selection)
- Produce testable predictions that can guide future observational programs

ASTRA's role in science is to work alongside the experienced astronomer to analyze data and facilitate genuine discovery. ASTRA is therefore a tool to assist the astronomer, rather than a replacement for domain expertise. The human scientist brings deep physical understanding, creative insight, and judgment about which questions are worth asking. ASTRA brings automated pattern discovery, causal reasoning, and systematic hypothesis evaluation. Together, they can pursue scientific questions more effectively than either could alone.

This collaborative model—ASTRA as discovery assistant working alongside domain experts—is the appropriate way to frame AI-assisted scientific discovery. Claims that AI systems will autonomously resolve major open questions in physics

overlook the essential role of human physical insight and the importance of careful experimental design. ASTRA's architecture supports discovery-mode operation, but definitive scientific discovery requires collaboration with domain experts and observational validation.

9.4 Limitations and Future Work

Current limitations include:

- **Sample sizes**: The test cases use 24-1,000 objects, which are small for production analyses. Scaling to larger datasets requires further validation.
- **Computational cost**: Causal discovery scales as $O(p^2)$ for p variables, limiting applicability to high-dimensional problems.
- **Prior dependence**: Bayesian evidence computation requires careful prior specification, and results can be prior-dependent.
- **Domain knowledge**: ASTRA's pattern recognition retrieves encoded domain knowledge. Genuine novel discovery requires identifying patterns not already in the knowledge base.
- **Selection effects**: Multi-wavelength cross-matching assumptions (Gaussian errors, uniform priors) may not hold in all regimes.

Future work should focus on: scaling to larger datasets; validation on more scientifically challenging problems; integration with time-domain surveys for real-time analysis; extension to gravitational wave and neutrino multi-messenger astronomy; and applying knowledge-isolation discovery mode to real datasets where causal structure is not known a priori.

10 CONCLUSION

We have presented ASTRA (Autonomous System for Scientific Discovery in Astrophysics), an integrated framework for physics-aware astrophysical analysis. Through six test cases, we have demonstrated ASTRA's technical capabilities while acknowledging the limitations of these demonstration cases.

Test 1: Scaling Relations Analysis (24 Herschel filaments)

- Identified universal width: 0.098 ± 0.019 pc
- Measured virial scaling with $r = 0.988$, $p < 10^{-18}$
- Found 2.7σ tension with theoretical prediction, warranting further investigation

Test 2: Multi-Wavelength Data Fusion (CDFS, 60/370 sources)

- Bayesian cross-matching with probabilistic associations
- Classified 41 AGN, 19 stars, 0 galaxies (selection effect discussed)
- Proper astrometric uncertainty propagation

Test 3: Pattern Recognition (600 SDSS galaxies)

- Identified established astrophysical relationships: mass-metallicity relation, environmental quenching
- Demonstrated pattern recognition capability retrieving known literature results
- Not novel discovery but validation of pattern recognition implementation

Test 4: Causal Inference (1,000 Gaia stars)

- Correctly recovered textbook causal relationships (validation demonstrator)
- Distinguished physical laws from selection biases
- Acknowledged this is a basic demonstration, not novel discovery

Test 5: Bayesian Model Selection (24 filaments)

- Power law and logarithmic models statistically indistinguishable (Bayes factor 1.2)
- Both strongly preferred over linear and broken power-law models
- Detected 2.7σ tension with virial theorem prediction
- Weakly-informative priors specified for reproducibility

Test 6: Discovery-Mode Operation (3,000 synthetic supernovae)

- Recovered embedded Phillips relation (decline rate–luminosity correlation) with $r = -0.91$, $p < 10^{-50}$
- Established causal structure through temporal analysis and counterfactual testing
- Triangulated evidence from numerical correlations, literature, visualization, and code
- Identified high-priority targets: super-Chandrasekhar SNe Ia, early-time excess events
- Demonstrated discovery-mode capability on synthetic data with known ground truth

Two Modes of Operation: Tests 1–5 validate ASTRA's ability to correctly recover known astrophysical results using real observational data. Test 6 demonstrates ASTRA's discovery-mode operation on synthetic data with embedded relations, validating that the discovery architecture can detect known patterns without prior specification.

ASTRA provides an integrated framework where physical validation, uncertainty quantification, causal reasoning, and Bayesian model selection are applied consistently throughout multi-step analyses. While individual components use established algorithms, their integration within a single framework enables analyses that would otherwise require manual combination of multiple separate tools and pipelines.

The complete source code, documentation, and reproducible analysis notebooks are available in our public GitHub repository White & Dey (2026). We encourage community feedback, contribution, and validation of these results on larger and more diverse datasets.

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AUTHOR CONTRIBUTIONS

Large language model assistance (Anthropic Claude) was used in manuscript preparation of this complex paper, in-

cluding drafting initial ideas, editing, and multiple rounds of peer review simulation. The scientific content, analysis, and conclusions were designed, executed, and verified by the authors. All errors and omissions are the responsibility of the authors. The Authors declare no conflict of interest.

DATA AVAILABILITY

All data and analysis scripts used in this paper are available to download directly from the GitHub repository White & Dey (2026). For access to updated versions of ASTRA, please contact the authors.

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